

Large Multimodal Model Clinical Diagnosis Support & Reasoning System for Tropical Diseases

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Abstract—Large multimodal models can reason over medical text and images, but they are often unreliable for tropical disease diagnosis, especially when using compact models in low-resource settings. In this work, we study whether retrieval-augmented generation and parameter-efficient fine-tuning can improve diagnostic reasoning for these models. We evaluate a lightweight multimodal framework on curated tropical disease cases from standard textbooks. Our results show that fine-tuning improves diagnostic accuracy and reasoning, while retrieval mainly helps with evidence grounding. This highlights both the potential and limits of retrieval-augmented multimodal systems for clinical support.

I. INTRODUCTION

Tropical and infectious diseases are often difficult to diagnose in low-resource settings due to limited access to specialist expertise and diagnostic tools. As a result, delayed or incorrect diagnoses remain common.

Recent language and vision–language models show potential for clinical decision support, but their reliability drops in specialized domains such as tropical diseases, especially for compact models suited to deployment in resource-constrained environments.

In this work, we examine whether retrieval-augmented generation (RAG) and parameter-efficient fine-tuning can improve diagnostic reliability for compact multimodal models. We propose a lightweight framework that combines external clinical case retrieval with supervised fine-tuning and evaluate it on curated textual and multimodal cases from tropical medicine textbooks.

Contributions. The main contributions of this work are:

- A lightweight multimodal diagnostic framework designed for tropical diseases that integrates retrieval augmentation and parameter-efficient fine-tuning.
- A curated and validated dataset of tropical disease clinical cases and synthetic diagnostic question–answer pairs.
- An evaluation of diagnostic reasoning, faithfulness, and retrieval quality using established clinical RAG evaluation protocols.

II. RELATED WORK

Recent studies show that large language models and multimodal models can support medical diagnosis, but their reliability decreases in specialized domains such as infectious and tropical diseases, where hallucination and weak reasoning are common [1].

Retrieval-augmented generation (RAG) has been proposed to improve factual grounding in clinical applications. Prior work demonstrates that multimodal retrieval can enhance reasoning and reduce hallucinations [2], [3], but most systems rely on large models or high-resource settings.

Other studies focus on evaluating clinical reasoning and improving model adaptation. MedCaseReasoning [4] provides structured evaluation protocols, while parameter-efficient fine-tuning has been shown to improve medical reasoning without full model retraining [5].

In contrast, our work targets compact multimodal models and examines how retrieval augmentation and parameter-efficient fine-tuning interact in tropical disease diagnosis.

III. METHODOLOGY

A. Task Definition

According to Dr Khanh, we study clinical diagnosis support for tropical diseases using textual and multimodal clinical cases. Given a case description, with optional medical images, the model is required to predict the most likely diagnosis and provide a open-ended clinical reasoning explanation. Performance is evaluated in both text-only and multimodal settings, focusing on diagnostic correctness, reasoning completeness, and evidence grounding.

B. System Overview

Our framework consists of three components: a compact multimodal vision–language model, an optional retrieval-augmented generation (RAG) module, and an optional parameter-efficient fine-tuning stage. Depending on the configuration, the model produces diagnostic predictions either directly, after fine-tuning, with retrieved clinical context, or using both fine-tuning and retrieval.

C. Retrieval-Augmented Inference

For retrieval-augmented settings, external clinical evidence is retrieved from a corpus of structured tropical disease cases compiled from standard textbooks. Text embeddings are generated using Bio ClinicalBERT [6], and image embeddings using CLIP-ViT-Base-Patch32 trained with natural language supervision [7]. Retrieved cases are reranked with a cross-encoder and appended to the input prompt to support evidence-grounded reasoning and reduce hallucinations.

D. Supervised Fine-Tuning

To improve domain-specific reasoning, we apply parameter-efficient supervised fine-tuning using QLoRA with 4-bit quantization. The training data consists of synthetic diagnostic question–answer pairs generated from curated tropical disease cases using a reasoning-aware prompt design. Fine-tuning aims to help the model internalize common diagnostic patterns, allowing robust reasoning even without retrieval. Models are evaluated both with and without retrieval to study how learned knowledge interacts with external evidence.

IV. DATA COLLECTION AND PREPARATION

A. Case Collection

Clinical cases were collected from three authoritative tropical medicine textbooks: *Clinical Cases in Tropical Medicine* [8], *Hunter’s Tropical Medicine and Emerging*

Infectious Diseases [9], and *The Diagnostics and Treatment of Tropical Diseases* by E. R. Stitt [10]. A total of 93, 67, and 30 cases were obtained from these sources, respectively.

B. Case Extraction and Structuring

Relevant cases were manually selected and converted into a standardized JSON format. Text extraction was performed using prompted inference with Gemini-2.5-Flash, while clinical images were extracted using PyMuPDF. Each structured case included patient demographics, clinical history, physical examination findings, laboratory results, imaging descriptions, and confirmed diagnoses.

C. Synthetic Question–Answer Transformation

Synthetic diagnostic question–answer pairs were generated from the structured cases using Gemini-2.5-Flash, following prompts adapted from the MedCaseReasoning framework [4]. The generated questions incorporated realistic clinical attributes such as age, sex, risk factors, and epidemiological context by following prompt from prior work on contextual tropical disease evaluation [11].

To encourage structured diagnostic reasoning during data synthesis, we adopted a reasoning-aware prompt design inspired by prior work on clinical reasoning with large language models [12]. Specifically, two illustrative few-shot examples from this framework were used as prompt demonstrations to guide the generation of step-by-step diagnostic rationales. These examples were used exclusively for data generation and did not overlap with any training or evaluation cases.

Approximately 400 QA pairs derived from 67 cases were used for supervised fine-tuning. For evaluation, 30 textual and 23 multimodal QA pairs derived from Stitt’s textbook were reserved as a held-out test set.

Full prompt templates used for case extraction and synthetic question–answer generation are provided in the Supplementary Material.

D. Validation and Quality Control

A two-stage validation process was applied to ensure both source faithfulness and clinical reasoning quality.

Stage 1: Extraction Fidelity Validation. The first stage verified that extracted clinical cases were semantically faithful to their original source texts. This step was motivated by the observation that hallucinations and omissions in medical text extraction often occur at the sentence or paragraph level, such as altered dosages, missing laboratory values, or invented clinical facts.

Each extracted textual block was compared against the most relevant segments of the original PDF using Sentence-BERT, a model optimized for sentence-level semantic similarity in clinical language. Cosine similarity scores were used to assess semantic overlap, and extracted content without a strong correspondence to the source was flagged as incomplete or potentially hallucinated. Only extractions that passed this semantic completeness check were retained for downstream processing.

Stage 2: QA Fidelity and Quality Validation. In the second stage, generated diagnostic QA pairs were evaluated using MedCaseReasoning-based prompts to assess (i) fidelity to the extracted case content, (ii) clinical reasoning soundness, and (iii) the absence of diagnosis leakage or unsupported inference.

For the held-out test dataset, 6 out of 30 QA pairs required manual review, with all failures originating from fidelity mismatches caused by outdated or inconsistent source references. These cases were corrected using updated medical sources, and no samples were discarded without verification.

For the fine-tuning dataset, 286 out of 429 QA pairs (66.7%) were retained. Excluded samples were removed due to critical medical errors, violations of source constraints, logical inconsistencies, or insufficient pedagogical quality, such as missing differential diagnoses or incomplete clinical information. This filtering ensured that the final datasets preserved source fidelity, clinical safety, and high-quality diagnostic reasoning.

V. MODEL BUILDING

Two vision-language models were selected as baselines: MedGemma-4B and LLaVA-NeXT, where LLaVA-NeXT [13] where LLaVA-NeXT extends the original LLaVA architecture with improved multimodal reasoning, OCR capability, and broader world knowledge. These models were chosen to represent compact and moderately sized multimodal architectures suitable for efficient deployment.

A. Retrieval-Augmented Generation

Text embeddings were generated using Bio_ClinicalBERT, while image embeddings were obtained using CLIP-ViT-Base-Patch32. Retrieved cases were reranked using a cross-encoder (cross-encoder/ms-marco-MiniLM-L6-v2) and indexed with FAISS. Retrieved clinical cases were appended to model prompts to support diagnostic reasoning.

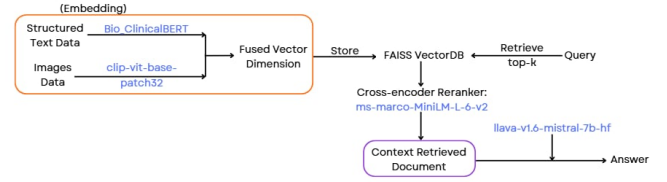


Fig. 1. Overview of the retrieval-augmented generation (RAG) pipeline for multimodal tropical disease diagnosis.

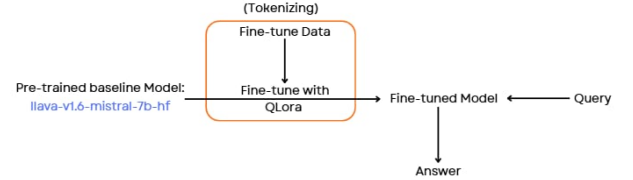


Fig. 2. Parameter-efficient fine-tuning pipeline using synthetic diagnostic question-answer pairs.

Figure 1 illustrates the retrieval-augmented generation (RAG) pipeline used in this study. Textual and visual queries are embedded, retrieved from the vector database, and appended as external context to support diagnostic reasoning.

B. Supervised Fine-Tuning

Parameter-efficient fine-tuning was performed using PEFT with QLoRA and 4-bit quantization. Approximately 300 QA pairs were used to adapt each model to tropical disease diagnostic reasoning. Models were evaluated both with and without retrieval augmentation.

Figure 2 shows the parameter-efficient supervised fine-tuning pipeline used to adapt the base multimodal models for tropical disease diagnostic reasoning.

VI. EVALUATION

Evaluation was conducted on a held-out test set derived from *The Diagnostics and Treatment of Tropical Diseases* by E. R. Stitt. While 30 test cases were prepared in total, only 23 cases contained both textual descriptions and associated clinical images. As a result, all multimodal experiments and reported results in this study are based on these 23 multimodal cases.

Two complementary evaluation paradigms were adopted. First, diagnostic quality was assessed using an LLM-as-a-judge approach with Gemini-2.5-Flash, following the evaluation protocols proposed in MedCaseReasoning and MetaRAG. This evaluation measured (i) diagnostic correctness, (ii) reasoning recall, and (iii)

faithfulness, defined as the degree to which the generated answer is supported by retrieved external evidence.

Second, retrieval quality was evaluated automatically using Precision@5, Recall, and Mean Average Precision (MAP), following the methodology of Alzheimer-RAG. These metrics were computed only for retrieval-augmented configurations.

Faithfulness was evaluated exclusively for retrieval-augmented models. For baseline prompting and fine-tuned models without retrieval, no explicit external context was provided at inference time, rendering faithfulness with respect to supporting evidence undefined. Accordingly, faithfulness scores for non-RAG configurations are reported as NA.

VII. EXPERIMENTS AND RESULTS

Experiments were conducted using the 23 multimodal test cases described in the Evaluation section. We evaluated six model configurations spanning baseline prompting, supervised fine-tuning, retrieval-augmented generation (RAG), and their combinations. Two vision-language models were considered: LLaVA-1.6-7B and MedGemma-4B. Each configuration produced both a predicted diagnosis and an accompanying clinical reasoning trace.

Table I reports the performance of all evaluated methods on the multimodal test set. Diagnostic correctness, reasoning recall, and faithfulness were assessed using an LLM-as-a-judge (Gemini-2.5-Flash). Automatic retrieval metrics were computed only for retrieval-augmented configurations.

Overall, the results indicate that supervised fine-tuning substantially improves diagnostic correctness and reasoning recall, while retrieval augmentation primarily contributes to external evidence grounding. However, the combination of fine-tuning and retrieval does not consistently yield higher faithfulness scores. This suggests that when strong internalized clinical knowledge is learned during fine-tuning, the model may rely less on retrieved context during inference, particularly when the retrieved cases are weakly aligned or incomplete.

The observed reduction in faithfulness for the Fine-tuned + RAG configuration therefore reflects a mismatch between retrieved evidence and the model’s internal reasoning, rather than increased hallucination. This finding highlights a key limitation of retrieval-augmented systems in specialized domains: retrieval corpus quality and relevance are critical for effective grounding, especially when models already possess strong domain priors.

VIII. DISCUSSION

Our results show that supervised fine-tuning is the main driver of improved diagnostic accuracy and reasoning quality for compact multimodal models. With domain-specific training, these models can learn meaningful diagnostic patterns for tropical diseases, even with limited capacity.

Retrieval augmentation mainly improves evidence grounding rather than diagnostic correctness. While RAG can support faithfulness by providing external references, its effectiveness depends heavily on the quality and relevance of the retrieved cases. When retrieval is weakly aligned, it can interfere with model reasoning instead of helping it.

The reduced faithfulness observed in the fine-tuned + RAG setting reflects a mismatch between internalized knowledge and retrieved evidence, rather than increased hallucination. This highlights a key limitation of RAG in specialized medical domains: retrieval quality is as important as the model itself.

Overall, these findings suggest that compact multimodal systems are viable for clinical decision support in low-resource settings, but careful data curation and retrieval design are critical for reliable grounding.

IX. CONCLUSION

In this project, we developed a compact retrieval-augmented multimodal clinical diagnosis and reasoning system tailored for tropical diseases. By combining parameter-efficient fine-tuning with external knowledge retrieval, we evaluated the trade-offs between internalized clinical reasoning and evidence-grounded generation in resource-constrained settings.

Our experiments on curated multimodal tropical disease cases show that supervised fine-tuning yields the most consistent improvements in diagnostic correctness and reasoning recall, while retrieval augmentation primarily contributes to external grounding and faithfulness. However, the interaction between fine-tuning and retrieval is non-trivial, highlighting the importance of retrieval corpus quality and alignment in specialized medical domains.

Overall, this work demonstrates both the potential and the challenges of deploying compact multimodal reasoning systems for clinical decision support in low-resource environments.

X. LIMITATIONS

This study has several limitations. First, the multimodal evaluation was conducted on a relatively small

TABLE I
PERFORMANCE COMPARISON ACROSS BASELINE WITH PROMPTING, FINE-TUNED, AND RETRIEVAL-AUGMENTED MODELS ON 23
MULTIMODAL TEST CASES.

Method	Model	Expected Output	LLM-as-a-Judge (Gemini-2.5-Flash)			Automatic Retrieval Metrics		
			Diagnosis	Reasoning	Faithfulness	Precision@5	Recall	MAP
Baseline	LLaVA-1.6-7B	Answer + Reasoning	0.522	0.529	NA	NA	NA	NA
	MedGemma-4B	Answer + Reasoning	0.522	0.626	NA	NA	NA	NA
Fine-tuned	LLaVA-1.6-7B	Answer + Reasoning	0.652	0.675	NA	NA	NA	NA
RAG	LLaVA-1.6-7B	Answer + Reasoning	0.174	0.509	0.705	0.574	0.870	0.636
	MedGemma-4B	Answer + Reasoning	0.478	0.492	0.458	0.600	0.826	0.673
Fine-tuned + RAG	LLaVA-1.6-7B	Answer + Reasoning	0.522	0.602	0.372	0.600	0.826	0.673

Diagnosis denotes diagnostic correctness; *Reasoning* denotes reasoning recall (coverage of clinically relevant reasoning steps); *Faithfulness* denotes the degree to which model outputs are supported by retrieved external evidence. Faithfulness is reported as NA for non-RAG configurations where no external context is provided.

test set of 23 cases, which may limit statistical robustness. Second, faithfulness scores for retrieval-augmented models were sensitive to the quality and relevance of the retrieval corpus, indicating that retrieval effectiveness alone is insufficient without well-aligned clinical evidence. Third, diagnostic evaluation relied on an LLM-as-a-judge framework, which, while standardized, may introduce subjective bias. Future work will focus on expanding the multimodal dataset, improving retrieval corpus curation, and incorporating clinician-in-the-loop evaluation.

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APPENDIX
SUPPLEMENTARY MATERIAL

PROMPT A: EXTRACTING TROPICAL BIG-CHUNK CASES IN BOOK INTO STRUCTURED JSON

```
"""# TropID Big-Chunk Case into JSON Extractor (Prompt)
## System
You are **TropID-Extractor**, an expert clinical information extractor for **tropical &
infectious diseases**.
Your task: **read a free-text clinical case** and return a **single JSON object** where **each
section is one coherent full-text block** (no bulletizing into tiny subfields).
If a section is not present, set it to 'null'. **Do not guess.**
**Output ONLY valid JSON** no preamble, no commentary.
### Formatting & Safety Rules
- **One JSON object only.**
- **Full-text blocks:** Each field below must be a **cohesive paragraph** (or short multi-
sentence block) stitched from the case text; paraphrase minimally, preserve clinical
meaning, and **do not invent** missing details.
- **Attribution discipline:** Prefer exact phrases from the source for key facts (fever pattern
, exposures, test names, titers) but keep the prose readable.
- **Units & names:** Keep units and proper names as written (C/F, NS1, thick smear, RDT,
species, titers/CT values).
- **Privacy:** Exclude any direct identifiers if present.
- **Uncertainty:** If the case explicitly says something is unclear/unknown, include that
wording.
- **Final diagnosis:** Write a concise paragraph that **states the diagnosis, the causative
agent if given, and the evidence** (labs/imaging/epidemiology/response to therapy).
- **Disease name (short):** After 'final_diagnosis', fill 'disease_name_short' with the **best
disease name only** (e.g., Dengue fever, Falciparum malaria, Scrub typhus).
---
Return **only** this JSON schema (exact keys, same order):
```json
{
 "patient_information": null,
 "chief_complaint": null,
 "history_of_present_illness": null,
 "exposure_and_epidemiology": null,
 "vitals": null,
 "physical_exam": null,
 "labs_and_diagnostics": null,
 "differential_diagnosis": null,
 "management_and_clinical_course": null,
 "final_diagnosis": null,
 "disease_name_short": null
}
```
### Field guidance (concise)
- **patient_information:** age/sex; relevant comorbidities/immunosuppression; vaccination/
allergy info if stated.
- **chief_complaint:** one-line problem + duration.
- **history_of_present_illness:** timeline, key symptoms, pertinent negatives, severity pattern
.
- **exposure_and_epidemiology:** residence/travel (place/setting, dates if present), vectors (
mosquito/tick), animals, water/food risks, contacts, season/outbreak context, occupation.
- **vitals:** all reported vital signs as text (fever values, BP, HR, RR, SpO).
- **physical_exam:** salient systems (skin, HEENT, chest, abdo, neuro, lymph, etc.).
- **labs_and_diagnostics:** CBC trends, key chem/coag, inflammatory markers, microbiology/
serology/PCR (assay + result + titer/CT if present), malaria tests, imaging highlights.
- **differential_diagnosis:** succinct narrative of the main alternatives considered, with one-
sentence justification for/against each.
- **management_and_clinical_course:** antimicrobials (drug/dose if given), supportive care,
procedures; response, complications, outcome.
```

```

- **final_diagnosis**: 3 to 5 sentences: explicit disease name causative agent, confirmation
  method (e.g., NS1+, thick smear species, PCR/serology), and why alternatives were ruled out
  .
- **disease_name_short**: the disease name only (no agent, no method).
---
## Few-shot (mini)
**Input (excerpt)**
32-year-old woman returned from rural Thailand 7 days ago. High fever to 39.5 C, severe
  myalgias, retro-orbital pain, and maculopapular rash. Platelets 7210^9/L, leukopenia.
  Dengue NS1 antigen positive on day 2. Managed with fluids and acetaminophen; no antibiotics
  . Recovered by day 6.
**Expected Output**
```json
{
 "patient_information": "32-year-old woman with no reported comorbidities or immunosuppression
 .",
 "chief_complaint": "Acute high fever with rash for approximately one week.",
 "history_of_present_illness": "Abrupt febrile illness reaching 39.5 C with severe myalgias
 and retro-orbital pain, followed by a maculopapular rash. Symptoms began shortly after
 travel and evolved over 7 days.",
 "exposure_and_epidemiology": "Recent travel to rural Thailand one week prior; mosquito
 exposure is implied by the endemic setting. No animal or water/food exposures or sick
 contacts were reported.",
 "vitals": "Maximum recorded temperature 39.5 C; other vital signs not reported.",
 "physical_exam": "Maculopapular rash and retro-orbital discomfort; no focal neurologic
 deficits or respiratory distress described.",
 "labs_and_diagnostics": "CBC notable for thrombocytopenia (platelets 7210^9/L) and leukopenia
 . Dengue NS1 antigen positive on day 2. No imaging was reported.",
 "differential_diagnosis": "Considered dengue fever versus chikungunya and malaria. Dengue
 supported by NS1 positivity, thrombocytopenia, and travel to an endemic area. Malaria less
 likely given dengue confirmation and no malaria-specific findings reported.",
 "management_and_clinical_course": "Supportive care with oral/IV fluids and acetaminophen; no
 antibiotics. Clinical improvement with full recovery by day 6; no complications documented
 .",
 "final_diagnosis": "Dengue fever due to dengue virus, confirmed by a positive NS1 antigen
 test. The diagnosis aligns with acute febrile illness, thrombocytopenia, retro-orbital pain
 , and compatible travel exposure. Malaria and chikungunya were considered less likely given
 confirmatory testing and symptom pattern.",
 "disease_name_short": "Dengue fever"
}
```
---
## User
Extract the following fields as **full-text blocks** from this clinical case:
"""

```

PROMPT B: IDENTIFY DISEASE, GENERATE AND SYNTHETIC DIAGNOSTIC QA

```
#   ### TASK: IDENTIFY DISEASES AND GENERATE Q&A
1. Identify EVERY distinct disease or pathogen mentioned in the source data.
2. For EACH identified disease, you MUST generate 4-6 unique diagnostic cases.

### TASK 1: DEMOGRAPHIC_SAMPLER_PROMPT
Generate a realistic patient DEMOGRAPHIC description based on one or more of the following:
ORIGIN, LOCATION, ETHNICITY, SEX or GENDER, AGE GROUP, SEXUAL ORIENTATION, SOCIOECONOMIC
STATUS, and DISABILITY STATUS.
- ORIGIN: Specific regions where the disease is highly prevalent.
- LOCATION: Specific towns or provinces within the selected ORIGIN where the disease is
highly prevalent.
- ETHNICITY: Local population at the selected ORIGIN or likely travelers. Specific as
possible.
- SEX: Male, female, or intersex.
- GENDER: Cis men, cis women, trans men, trans women, non-binary people.
- AGE GROUP: Young, elderly, child, adolescent, middle-aged, adult. For WOMEN also consider
pre-menopausal, post-menopausal.
- SEXUAL ORIENTATION: Straight, gay, bisexual, pansexual, asexual, queer.
- SOCIOECONOMIC STATUS: Low-income, middle-class, high-income.
- DISABILITY STATUS: Able-bodied, autistic, deaf, blind, deaf-blind, hearing impairment,
intellectual disability, orthopedic impairment, learning disability, speech or language
impairment, traumatic brain injury, visual impairment.
- RULES: Do NOT make any claims about the demographic. Do NOT output a sentence. Prioritize
ORIGIN and LOCATION. Do NOT mention medical conditions. Adjust SOCIOECONOMIC STATUS to
match the ORIGIN.

### TASK 2: STRUCTURED OUTPUT (XML TAGS)
For each case, use these tags:
1. <think>
  - Key points: What makes this case pedagogically interesting?
  - Analytic distinctions: How do you separate the final diagnosis from look-alikes?
</think>
2. <case_prompt>
  - DO NOT use a list or header for demographics.
  - Start directly with the patient's presentation as a narrative.
  - Weave the ORIGIN, LOCATION, ETHNICITY, SEX, AGE, and SES naturally into the first two
sentences of the story.
  - Continue with the history of present illness, symptoms, and clinical findings in a
professional paragraph-style narrative.
  - End with a varied diagnostic question.
  - DO NOT provide multiple-choice options (A, B, C).
</case_prompt>
3. <diagnostic_reasoning>
  - Write as a narrative sequence of COMPLETE SENTENCES.
  - Use a logical Chain-of-Thought (CoT) connecting findings to pathology.
  - Do NOT use bullet points.
</diagnostic_reasoning>
4. <final_diagnosis>
  - The single disease name only.
</final_diagnosis>

### FINAL JSON OUTPUT FORMAT:
Each object in the returned JSON list MUST include:
"source_file": "{filename}",
"think": "Contents of <think>",
"question": "Contents of <case_prompt>",
"diagnostic_reasoning": "Contents of <diagnostic_reasoning>",
"answer_diagnosis": "Contents of <final_diagnosis>"
"""
```